

30 August 2026

The Editors JMIR Mental Health

Dear Editor,

We are pleased to submit our original research manuscript, **“Typed Feature Vectors, Generalized Pretrained Transformer Embeddings of Deterministic Patient Narratives, and Nearest-Neighbor Retrieval for Predicting a Treatment-Switch-Defined Electronic Health Record Proxy for Treatment-Resistant Depression: Retrospective Cohort Study,”** for consideration in *JMIR Mental Health*.

This study asks whether routinely collected electronic health record (EHR) data can predict a treatment-switch-defined EHR proxy for treatment-resistant depression (TRD) at the point of a patient’s index antidepressant prescription. It makes two points, and the manuscript is organized around them.

The first is that writing the record out as text and embedding it with a general-purpose transformer does not beat a transparent typed feature vector built from the same record. In a cohort of 42,579 patients the paired difference between the two best models is +0.008 ROC AUC (95% CI −0.003 to +0.019), and the result holds across four encoders. The comparison is nevertheless not a simple tie: the embedding significantly helps logistic regression and significantly hurts all three tree ensembles, so representation and learner are one design choice rather than two.

The second is that predicting a patient from their nearest neighbors in that embedding — the clinical digital-twin premise — recovers real signal and still loses decisively to a model fitted on the same representation. Nearest retrieval reaches 0.594 against 0.499 for random retrieval and 0.432 for farthest retrieval, which establishes that proximity in the embedding tracks TRD risk. It nevertheless falls 0.063 ROC AUC short of the trained classifier, on intervals that do not overlap, and that shortfall reproduces on every encoder tested.

One thing is true of both results and bounds both, and the manuscript stresses it rather than leaving it to be inferred: in both arms the patient data were hand-picked. Predictor selection ran once, before either representation existed, and the narrative the encoder reads is a fixed template over that same selection. Serialize-and-embed is often recommended as a way to stop choosing features, and a reader arriving at a parity result with that expectation will read it as “the embedder did as well with no feature engineering”. The engineering was relocated, not removed, and a whole untailored record was never embedded. Supplement S10 is the field-by-field evidence.

Discrimination is modest throughout (ROC AUC $\approx$ 0.65) and the predictive signal localizes to clinical content, psychiatric history and medication burden, rather than to the sociodemographic fields supplied to the models. Subgroup discrimination and calibration are reported across eight strata with multiplicity control. We are explicit throughout that the target is a treatment-switch proxy rather than measured non-response, that the study is an internal validation by a single random split, and that it reports no fairness assessment. A Limitations section states these and seven others in order of how much they bound the conclusions.

This work is reported in accordance with the **TRIPOD+AI** guideline for the reporting of clinical prediction models that use regression or machine learning methods; a completed TRIPOD+AI checklist is included with this submission.

We confirm that this manuscript describes original work, is not under consideration for publication elsewhere, and that all listed authors have read and approved the submission. The EHR data are not publicly shareable; the full analysis code is available at <https://github.com/Pirate-Hunter-Zoro/TRD-EHR>. Funding, conflict of interest, and ethics/IRB statements are provided in the manuscript's Declarations: no conflicts of interest are declared; the work was funded by the William K. Warren Foundation, with no other funding source; and the study is a secondary analysis of de-identified data and is therefore not human-subjects research subject to institutional review board approval. All analysis was performed on local institutional hardware with a de-identified extract containing no protected health information, with no patient data transmitted to any external or commercial service.

Thank you for considering our work.

Sincerely,

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